Assignment 02 - COAL

Date:

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Q1.

```
mov ax, 3        ; ax = 3
mov bx, ax       ; bx = 3

shl ax, 4        ; ax = 3 * 16 = 48
mov cx, bx       ; cx = 3
shl cx, 3        ; cx = 3*8 = 24
add ax, cx       ; ax = 48 + 24 = 72
mov cx, bx       ; cx = 3
shl cx, 1        ; cx = 3*2 = 6
add ax, cx       ; ax = 72 + 6 = 78
```

Q2.

```
mov eax, 02345ABCDh

ror eax, 8
ror eax, 8
ror eax, 8
ror eax, 8

rol eax, 8
rol eax, 8
rol eax, 8
rol eax, 8
```


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Q3. mov eax, 02345ABCDh

```
ror eax, 16
ror eax, 16

rol eax, 16
rol eax, 16
```

Q4. DX = 0101100111011010

a)
```
mov ax, dx
and ax, 001Fh ; Mask bits 0-4 (0000 0000 0001 1111)
mov BSecords, al
```

b)
```
mov ax, dx
and ax, 07E0h ; Mask bits 5-10 (0000 0111 1110 0000)
shr ax, 5
mov BMinutes, al
```

Q5. dx = 9A3Fh
ax = 7F80h
CF = 0
ax = 7F80h + 90EAh = 110GAh. ax = 106Ah, CF = 1
dx = 9A3Fh + 1 + 1 = 9A4[illegible]1h. dx = 9A4[illegible]h, CF = 0
CF = 1

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ax = 106Ah + 8120h = 918Ah . ax = 918Ah , CF = 0
dx = 9A41h + FFFFh + 0 = 19A10h . dx = 9A40h , CF = 1

dx = 9A40h , ax = 918Ah , CF = 1

Q6. a) test ax, ax
b) test bx, 1
c) test dx, dx
d) test dx, 8000h
e) test bl, bl

Q7. DX = 0005h
CF = 1
AX = 0010h
dx = 0005h + 0010h + 1 = 0016h

DX = 0016h

Q8.

Errors	Corrections
1. mov esi, val1	mov esi, OFFSET val1
2. dec esi	inc esi
3. dec edi	inc edi
4. mov BYTE PTR [esi], al	mov BYTE PTR [ebx], al
5. mov cx, 8	mov ecx, 8
6. mov edi, val2	mov edi, OFFSET val2
7. missing line	mov ebx, OFFSET result

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Q9.
```
.data
    inputMsg BYTE "Enter Student's Marks (0-100): ", 0
    gradeMsg BYTE "The Grade is: ", 0

.code
main PROC
    mov edx, OFFSET inputMsg
    call WriteString
    call ReadInt

    cmp eax, 80
    jae grade_A
    cmp eax, 60
    jae grade_B
    cmp eax, 40
    jae grade_C
    jmp grade_F

grade_A:
    mov al, 'A'
    jmp display

grade_B:
    mov al, 'B'
    jmp display
```

Date: _____

```
grade_C:
        mov al, 'C'
        jmp display

grade_F:
        mov al, 'F'

display:
        mov edx, OFFSET gradeMsg
        call WriteString
        call WriteChar
        call Crlf

        exit
main ENDP
END main
```

Q10.

eax = 3
ebx = 7
Stack = 3
Stack = 7, 3
eax = 7 , Stack = 3
Stack = 10, 3
ebx = 10 , Stack = 3
eax = 3 , Stack = -

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EAX = 3 , EBX = 10 , Stack = empty

Q11.

```
mov eax, num1
cmp eax, num2
jle check_else_if

mov eax, num3
cmp eax, num4
je check_else_if

mov eax, num1
add eax, num3
mov result, eax
jmp end_if

check_else_if:
        mov eax, num2
        cmp eax, num4
        jne else

        mov eax, num2
        sub eax, num1
        mov result, eax
        jmp end_if
```

Date:

```
else:
    mov result, 0

end_if:
```

Q12.

```
while_loop:
    mov eax, op1
    cmp eax, op2
    jg end_while

    mov eax, op1
    cmp eax, x
    jl else
    cmp eax, y
    jg else

    add z, 10
    jmp next

else:
    sub z, 10

next:
    dec op1
    jmp while_loop
```


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Date:

```
end_while:
        exit
```


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